

How the literature search helps your dissertation

Strands 1 and 2 of the LitReviewKit search (28 September 2026): what was found, what each paper says, and where it fits in “This Time Is Different?”

Prepared for Shahram · MSc Business Analytics, Dublin Business School · companion to the Literature Handbook v2 and the Chapter 2 Blueprint · updated 30 September 2026 (exploratory extensions added; Section 10 corrections applied)

Read this first
This is a study guide, not dissertation text. Do not copy its sentences into the dissertation. Read each paper, then write in your own words. The same rule as your AI-use log applies: AI helped plan and search; every sentence of the dissertation is yours.
Every paper note rests on the abstract or the title — not the full paper. Each card says which (see the key below). Before you cite a finding, confirm it in the paper itself, as you did for the Handbook cards.
The “with or against your results” notes are prompts for your Discussion, not conclusions. Where a paper suggests an explanation for one of your results (for example, the training loss), treat it as something to examine and argue, not as a proven cause.

Update — 30 September 2026
Two exploratory extensions were added after your main results (Amendment A1). E1 compares the SHAP shares with a Diebold–Yilmaz spillover measure; E2 checks the volatility paradox. Sections 4 and 5 now include them, and card B5 is updated.
Two papers the searches did not find must be cited: Feng, Shi & Kutan (2026) and Brini (2026) — see Section 8 and the new cards in the Literature Handbook v2.
The corrections in Section 10 have been applied to the Literature Handbook v2, the Chapter 2 Blueprint, the Research Plan and the Project Handbook.

Key	Meaning
[A]	The note is based on the paper's abstract (read from the search records).
[T]	No abstract was available: the note is based on the title and metadata only. Read the paper before citing anything specific.
[H]	Already in your Literature Handbook v2 (with page-level notes there). The card here adds how it relates to your results.
★★★	Read in full now — it changes how you write Chapter 2, 3 or 5.
★★	Read the abstract, introduction and main results table.
★	Skim; cite only if you need the point.

1. What the search gave you — in seven points

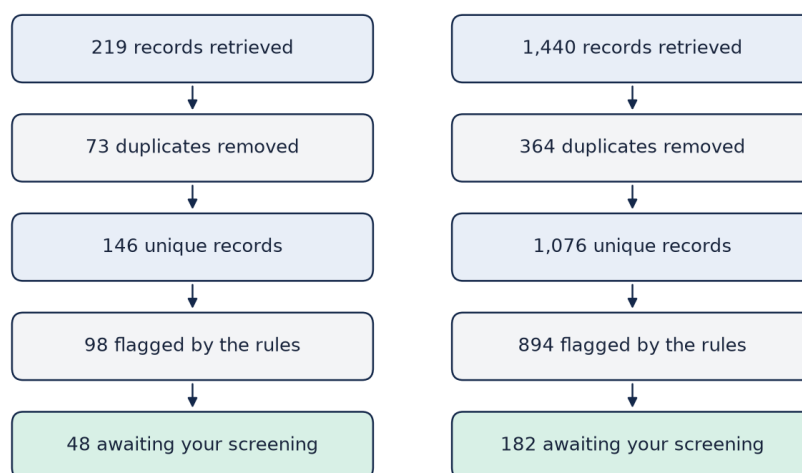
The two searches found about 55 papers worth your attention. About 45 of them are not yet in your Literature Handbook; about 10 are (Kearney 1998, Gallagher & Twomey 1998, Corbet & Twomey 2015, Kim & Won 2018, Bucci 2020, Christensen et al. 2023 and others). They help in seven concrete ways:

- **1. A proper Irish evidence base for Section 2.3.** New Ireland-specific studies: Choi & Murray (1995), Alles & Murray (2001), Bredin, Gavin & O'Reilly (2005) on US monetary news and Irish volatility, Sharkasi et al. (2006) on long memory in ISEQ volatility, and Mitra & Duggar (2007/2009) on contagion risk from the UK, US and Netherlands to Irish banks. Section 2.3 can now cite evidence about the ISEQ itself, not only about “Europe”.
- **2. Published company for your negative H1 result.** Audrino & Chassot (2025, International Journal of Forecasting) and Kılıç (2025, Federal Reserve discussion paper) both find carefully fitted HAR-type models beat machine learning; Rahimikia & Poon find ML wins on normal days but fails on jump days — your exact pattern (near the benchmarks in calm periods, far worse in the GFC). Your result is part of a live debate, not an anomaly.
- **3. Candidate explanations to examine in Chapter 5.** The papers point to specific differences between your set-up and studies where ML wins: training loss (Jia & Yang: MSE-type losses handicap deep models; you train on MSE and evaluate on QLIKE), few predictors (Kılıç; you have six), data size and pooling (Zhang et al.; you have one market), fitting scheme (Audrino & Chassot; you use expanding windows with annual refits) and horizon (Christensen et al.).
- **4. The closest precedent for your SHAP method — and a boundary for your novelty claim.** Choi, Shin & Shin (2024) propose “vector SHAP” and group foreign indices' contributions by **region** to explain Korea's realised volatility — very close to your channel grouping. Shin et al. (2025) apply it to an LSTM on S&P 500 realised volatility. You must cite and position against both (Section 5 of this guide).
- **5. Evidence for the gap.** Gunnarsson et al. (2024), a systematic review, states that very few volatility-forecasting papers use explainable AI. Neither search found a study that combines the Irish market, deep learning, SHAP and crisis-by-crisis comparison (a targeted check found two low-visibility items to rule out first — Section 8).
- **6. Material to interpret each H2 result.** Corbet & Twomey (contagion eased after the troika bail-out — most of your Irish window is after it), Carneiro et al. (2025: the only COVID contagion link into the ISEQ ran through France's CAC 40, while your DAX-based euro share fell), Goldstein et al. (2019: peripheral markets co-move most with the core — supports the DAX), Izzeldin et al. (2026: Brexit hit Ireland — supports H2d).
- **7. A documented, reproducible search** for your search-strategy paragraph (Chapter 2 or 3), with dates, sources, queries, counts and limitations (Section 9) — and one **correction** for your Handbook: the 1998 spillover paper is by Gallagher **and** Twomey.

2. What the search did

Tool: LitReviewKit, an open-source Streamlit app (github.com/Ulviyabasova/LitReviewKit, commit 5236316), run on your computer with three local changes: saved search presets for your two strands; Semantic Scholar results sorted by citation count and cut to the limit (its bulk search is unranked and returned up to 1,000 records per query); and the per-source cap split equally across queries. It searches the public Crossref, OpenAlex and Semantic Scholar databases — **not** Scopus or Web of Science (no institutional API key).

Strand 1 — Irish crisis transmission Strand 2 — ML vs GARCH/HAR + SHAP



Preparation counts from the app, 28 Sep 2026 — not a completed PRISMA flow. Screening, full-text reading and citation chasing are still to do.

Figure 1. Record counts at each automatic step. “Flagged by the rules” means the app’s keyword rules suggested exclusion; those records stay in the export and you can still include them.

	Strand 1 — Irish crisis transmission	Strand 2 — ML vs GARCH/HAR + SHAP
Question	Evidence on volatility, contagion and spillovers in the Irish market	Deep learning / ML vs GARCH and HAR for volatility forecasting, and SHAP explanations
Queries sent (9 each)	{ISEQ, Irish stock market, Irish equity} × {volatility, contagion, spillover}	{LSTM, machine learning, SHAP} × {realized volatility, realised volatility, volatility forecasting}
Extra screening terms	Concept A also: Ireland, Irish. Concept B also: spillovers, volatilities, co-movement, comovement, interdependence, GARCH, EGARCH, HAR	Concept A also: deep learning, long short-term memory, neural network(s), recurrent neural network, Shapley, explainable, XAI. Concept B also: volatility forecast(s), forecasting (the) volatility, volatility prediction, predicting (the) volatility, volatility modelling, conditional variance, GARCH, EGARCH, HAR model, HAR-RV, heterogeneous autoregressive
Exclusion terms	wind power, wind energy, electricity, dairy, milk, agricultural	bitcoin, cryptocurrency, cryptocurrencies, electricity
Field signals	stock(s), equity/equities, share prices, stock exchange, financial, bank(s), sovereign, bond(s)	none
Years	1995–2026	2010–2026
Sources that worked	Crossref (9/9 queries, 23 records); Semantic Scholar (8/9, 196). OpenAlex failed (its anonymous search was paused that day, HTTP 503).	Crossref (9/9, 4 records); OpenAlex (9/9, 900); Semantic Scholar (9/9, 536)
Counts	219 retrieved → 146 unique → 48 pass the rules	1,440 retrieved → 1,076 unique → 182 pass the rules
Main exclusion reasons (overlap)	below relevance threshold 47; missing both concepts 44; no field signal 38; missing Ireland concept 32; missing volatility concept 11	missing volatility concept 675; below threshold 342; exclusion term 152; missing both 152; missing ML concept 41
Notes	Re-run now that OpenAlex search is back, before you finalise the counts.	A first run was discarded: its rules let in about 200 stock-price papers and excluded Bucci (2020). Rules were revised and the search re-run. Record this honestly.

How the app scores relevance: for each concept group matched, +30 if in the title, +20 if in the abstract, +12 otherwise (key words); plus up to +10 for citations; papers under 30 are flagged. Records with no abstract can only match on the title, so **auto-exclusions need a human look**.

3. Where the papers go in your dissertation

Section numbers follow your Chapter 2 Blueprint. Card numbers (A1, C4 ...) point to Section 6 of this guide.

Section	Papers from this search	What they let you do
2.2 Why crises recur	Kelly & Ó Gráda 2000 (A11); Qiu et al. 2025 (D3)	Historical contagion through information networks; a history-of-methods spine
2.3 Three crises and Ireland	Kearney 1998 (A1); Gallagher & Twomey 1998 (A2); Choi & Murray 1995 (A3); Alles & Murray 2001 (A4); Bredin et al. 2005 (A5); Sharkasi et al. 2006 (A6); Mitra & Duggar 2007/09 (A7); Corbet & Twomey 2015 (A9); Conlon & Cotter 2014 (A12); Aslam et al. 2021 (B7); Carneiro et al. 2025 (B8); Izzeldin et al. 2026 (B10); Niu et al. 2023 (C20)	Irish stylised facts; the US and UK channels before 2008; the Irish banking and sovereign crisis; COVID, Brexit and 2022 evidence that includes Ireland
2.4 How shocks travel	Ahmad et al. 2014 (B1); Dajčman 2013 (B2); Trabelsi & Hmida 2018 (B3); Goldstein et al. 2019 (B4); Mensi et al. 2018 (B5); Caporale et al. 2016 (B6); Gołab & Zamojska 2019 (B9)	Applications of DCC-GARCH, Forbes–Rigobon and Diebold–Yilmaz that include Ireland
2.5 Forecasting volatility	C1–C23; reviews D1–D4	The ML vs econometrics debate on both sides; hybrids; data, loss and horizon differences
2.6 Explaining ML forecasts	Choi et al. 2024 (E1); Shin et al. 2025 (E2); Kang et al. 2026 (E3); Hashami & Maldonado 2025 (E4); Rahimikia & Poon (C3); Christensen et al. 2023 (C6); Gunnarsson et al. 2024 (D1)	Grouped (vector) SHAP; regime-dependent explanations; how rarely XAI is used
2.7 Gap and hypotheses	Gunnarsson et al. 2024 (D1); Ge et al. 2022 (D2); Choi et al. 2024 (E1); Shin et al. 2025 (E2)	State the gap with a review's evidence and position against the closest work
Chapter 3 Method	Audrino & Chassot 2025 (C1); Jia & Yang 2021 (C4); Taneva-Angelova & Granchev 2025 (C10); Moreno-Pino & Zohren 2024 (C16); Goldstein et al. 2019 (B4); Dajčman 2013 (B2); Choi et al. 2024 (E1); Kang et al. 2026 (E3)	Justify walk-forward refits, QLIKE + DM at 5 days, daily RV, DAX as the euro channel, fixed windows, grouping SHAP by channel and window
Chapter 5 Discussion	See Section 4 of this guide	Compare each hypothesis result with published evidence
Chapter 6 Limitations / future work	Kim & Won 2018 (C8); Kılıç 2025 (C2); Souto & Moradi 2023 (C12); Song et al. 2022 (C15); Chun et al. 2024 (C19); Zhang et al. 2023 (C7); Carneiro et al. 2025 (B8); Mensi et al. 2018 (B5)	Hybrid GARCH-LSTM, regime-switching HAR, turbulence input, macro inputs, economic value, pooling markets, CAC 40 proxy, Diebold–Yilmaz (PhD)

4. Your results against the literature

Your pre-registered results (README) set against what these papers report. Use it to plan Chapter 5 — then write it yourself after reading the papers.

Result	What you found	Papers in line with it	Papers against it	Points to examine
H1 LSTM beats GARCH and HAR	Not supported: QLIKE 0.463 vs 0.380 (GARCH) and 0.414 (HAR); LSTM significantly worse ($p = 0.003, 0.044$)	Audrino & Chassot C1; Kılıç C2; Rahimikia & Poon C3 (jump days); Gunnarsson D1 (econometric models often similar); Hansen & Lunde [H]	Bucci C5; Christensen C6; Zhang C7; Kim & Won C8 (hybrid); Liu C9; Taneva-Angelova C10; Chung Alva C11	Training loss (C4); six inputs (C2, C6); one market, ~1,000 days in the first window (C7, C22); expanding window + annual refit (C1); 5-day horizon (C6, C11)
H1 by window	LSTM best in COVID (0.562 vs 0.700) and 2022 (0.264 vs 0.290); worst in the GFC (1.021 vs 0.401)	Taneva-Angelova C10 and Ramos-Pérez C14 (DL better in volatile periods); Souto & Moradi C12	Rahimikia & Poon C3 (fails on jumps) explains the GFC	Why COVID but not the GFC: by 2020 the training data contained the GFC and the Irish crisis (links to H3)

Result	What you found	Papers in line with it	Papers against it	Points to examine
H2a GFC: US share rises most; domestic above calm	US rises (0.266 → 0.299); domestic falls (0.523 → 0.482)	Bredin A5 and Mitra & Duggar A7 (a real US channel); Trabelsi B3 (contagion everywhere in the subprime crisis)	Bekaert et al. [H] stress domestic factors	Direction share 0.62 in the GFC: asymmetry (Choi & Murray A3; Caporale B6)
H2b Irish crisis: euro and domestic rise; US below GFC	Euro rises (0.236); US below GFC (0.252); domestic does not rise (0.512)	Corbet & Twomey A9 (contagion eased after the Nov 2010 troika programme); Trabelsi B3 (weak sovereign-crisis contagion)	Ahmad B1, Dajčman B2, Mensi B5 (Ireland as a source of contagion)	Direction: those papers study Ireland → Europe; your SHAP shares show what the forecaster uses for Ireland. Lag profiles same as calm (slow-burning crisis)
H2c COVID: shares move towards equality	Narrowly supported (spread 0.308 vs 0.312)	Aslam B7 (78% of variance from spillovers; everything connected)	Carneiro B8 (contagion into the ISEQ via the CAC 40, while your DAX share fell to 0.180)	Is the DAX the right euro proxy? CAC 40 robustness idea
H2e 2022: euro share rises	Not supported (euro 0.211 → 0.143; US 0.317)	—	Boungou & Yatié [H]	Little Ireland-specific 2022 evidence found — read Niu C20; state the gap honestly
H2d Brexit (FTSE run)	Pending	Izzeldin B10 (Ireland particularly affected); Goldstein B4	Li (2020) [H]	Decide after the FTSE run
H3 Frozen-model degradation GFC > COVID > Irish	GFC 4.26 > Irish 1.33 > COVID 0.74 × HAR; GFC largest, as predicted	Rahimikia & Poon C3; Kılıç C2 (regimes); Chung Alva C11 (breaks)	—	Audrino & Chassot C1: training window drives performance — your frozen vs refitted test is a fitting-scheme experiment
H4 Stable rankings across seeds (≥ 0.8)	Not supported: 0.7 in the GFC and Irish windows; 1.0 in calm and 2022	Christensen C6 (models agree on important predictors but not their ranking); Choi E1 (sampling SHAP is sensitive)	—	Is it the model (seeds) or the explainer (sampling)? Vector SHAP as a remedy
Domestic channel largest everywhere	≈ 0.48–0.54 in every window	Shin E2 (own past RV dominates); Sharkasi A6 (long memory)	—	Expected for volatility; frame foreign shares as the informative part
E1 (exploratory) SHAP vs Diebold–Yilmaz spillovers	Weak agreement: 4 of 8 direction checks, 2 of 5 rankings. Both show the US share highest in COVID-19; in 2022 the spillover euro share rises (0.381) while SHAP's falls (0.143). Total spillover 34.0%; ISEQ a net receiver (−8.4)	Aslam B7 and Mensi B5 (spillovers intensify in crises: your total spillover rises from 32.6% in calm to 49.5% in COVID-19)	Mensi B5 finds Ireland a net transmitter (you find a net receiver — different markets and period)	The two measures are different objects (linear VAR on volatility vs SHAP on a non-linear model with return inputs); Carneiro B8 — DAX vs CAC 40 as the euro proxy
E2 (exploratory) volatility paradox	Only Germany unusually calm before the GFC (4th percentile); Ireland 40th, US 42nd; no market before COVID-19	Danielsson et al. (2018) — not from the searches (Handbook v2, D1) — for Germany	Same paper, for Ireland and the US	Monthly averages; one crisis per market; the Irish boom may show in credit (Kelly 2009 [H]) more than in equity volatility

5. Positioning your contribution after this search

Facts your gap statement (Section 2.7) must respect. Write the paragraph yourself.

Already done by others	Still distinctive in your study
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Already done by others	Still distinctive in your study
ML/DL vs GARCH/HAR for realised volatility — many studies, both outcomes (Part C)	The Irish market (ISEQ) as the target — no ML volatility study focused on the ISEQ appeared in either search (1,222 unique records). A targeted OpenAlex query (“ISEQ LSTM”, 24 hits) found only multi-market or low-visibility items — rule them out first (Section 8)
SHAP for realised-volatility models (Rahimikia & Poon C3; Shin et al. E2)	Explanations compared crisis by crisis against pre-registered predictions based on each crisis's origin (RQ2)
Grouping lagged SHAP values by variable and by region of foreign markets (Choi et al. E1)	Channel shares with block-bootstrap intervals and seed stability (RQ4)
Regime-dependent interpretability (Kang et al. E3; Hashami & Maldonado E4)	A frozen-model “learning from history” (Minsky) test across three crises (RQ3)
Contagion into and out of Ireland with DCC/DY/copulas (Part B)	A forecasting-and-explanation lens on transmission, reported as fingerprints, not causes
Sahiner (2023) and Ahmed (2026) — deep learning with spillovers/SHAP [H]	Negative results reported in full, as pre-registered
Spillover / connectedness indices for Irish and European markets (Mensi B5; Aslam B7; Gořab B9)	Using a spillover index as a second lens on a forecasting model’s explanations (exploratory E1) — check for prior work before claiming it; Feng et al. (2026) is close in spirit

6. The papers, one by one

Each card: what the paper does, what it finds (from the abstract), why it matters for your dissertation, where it goes, how it sits against your results, and what to check before citing. Numbers such as QLIKE values or channel shares are **your** results from the README.

Part A — The Irish market (Strand 1 core)

These are the Irish-specific studies the search surfaced. Together they give Section 2.3 an evidence base that is about the ISEQ itself, not just about Europe in general.

A1 Kearney (1998) — Causes of volatility in a small, internationally integrated stock market: Ireland

Journal of Financial Research 21(1) · doi 10.1111/j.1475-6803.1998.tb00271.x · Basis [A][H] · Priority ★★ · Strand 1 (Semantic Scholar) · 62 citations

What it does	Relates Irish stock-market conditional volatility (July 1975 – mid-1994) to UK market volatility and to business-cycle variables.
What it finds	Exchange-rate volatility matters more than interest-rate volatility for a small integrated market; the author suggests membership of the European Monetary System could lower volatility in small member states.
Why it matters to you	The classic statement that Irish volatility is imported : it justifies modelling ISEQ volatility with foreign-market inputs at all. It predates the euro (1999). Your sample starts in 2003, so the currency channel Kearney stresses is largely gone — one reason your channels are US and euro-area equity markets rather than exchange rates.
Where it goes	2.3.1 (Ireland as a small open, integrated market); 2.4 (transmission); Chapter 3 input choice.
Check before citing	Already carded in your Handbook v2. The search metadata dates it 1997; the journal volume is 1998 — cite 1998.

A2 Gallagher & Twomey (1998) — Identifying the source of mean and volatility spillovers in Irish equities: a multivariate GARCH analysis

Economic and Social Review (per your Handbook) · no DOI in metadata · Basis [T][H] · Priority ★★★ · Strand 1 (Semantic Scholar) · 39 citations

What it does	From the title: a multivariate GARCH study of where mean (return) and volatility spillovers into Irish equities come from.
Why it matters to you	Closest pre-euro ancestor of your RQ2: it asks which foreign market transmits into Irish returns and volatility. Your SHAP channel shares ask the same question with a different tool (an explained LSTM instead of a multivariate GARCH). Co-authored by Cian Twomey — your lead PhD supervisor target. Being able to say how your 2003–2023 fingerprints compare with his 1998 spillover evidence is a strong opening for that conversation.
Where it goes	2.3.1 and 2.4; Chapter 5 (compare your channel ranking with the pre-euro evidence).
Check before citing	Your Handbook v2 lists this as “Twomey, C. (1998)”. The search metadata gives two authors: Liam A. Gallagher and Cian Twomey . Confirm on the Economic and Social Review archive and correct the Handbook card, the Blueprint reference list and your reference manager. No abstract was available, so read the paper before describing its findings.

A3 Choi & Murray (1995) — Returns and volatility: a study of the Irish equity market

Accounting, Finance & Governance Review · doi 10.52399/001c.35411 · Basis [A] · Priority ★★ · Strand 1 (Semantic Scholar)

What it does	Tests two classic ideas on Irish data: (1) expected returns rise with expected volatility (a risk premium) and (2) unexpected returns move opposite to unexpected volatility. Expected volatility comes from ARMA models; conditional variances from several ARCH variants.
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Why it matters to you	Early evidence that ARCH-type models are appropriate for the Irish market — it supports GARCH(1,1) as a benchmark for the ISEQ , not only for large markets. Hypothesis (2) is the asymmetry idea (bad news raises volatility more). Your SHAP result that the direction of returns carries 62% of attribution in the GFC (41% in calm periods) is a modern echo of it.
Where it goes	2.5 (why GARCH for Ireland); Chapter 5 (direction vs size result).
Check before citing	The abstract does not state the results; read the paper before saying which hypothesis held.

A4 Alles & Murray (2001) — An examination of return and volatility patterns on the Irish equity market

Applied Financial Economics · doi 10.1080/096031001750071523 · Basis [T] · Priority ★★ · Strand 1 (Crossref + Semantic Scholar) · ABDC B

What it does	From the title: descriptive/econometric study of return and volatility patterns on the Irish market.
Why it matters to you	A second Irish stylised-facts source (with A3 and A6). Useful to show that volatility clustering and related patterns are documented for the ISEQ specifically — the premise of every model you fit.
Where it goes	2.3.1; 2.5 (stylised facts, next to Cont, 2001).
Check before citing	No abstract available — read before citing any specific finding.

A5 Bredin, Gavin & O'Reilly (2005) — US monetary policy announcements and Irish stock market volatility

Applied Financial Economics · doi 10.1080/09603100500390836 · Basis [T] · Priority ★★★ · Strand 1 (Crossref + S2) · ABDC B · 29 citations

What it does	From the title: studies how US monetary-policy announcements affect the volatility of the Irish stock market.
Why it matters to you	Direct pre-crisis evidence of a US channel into Irish volatility — exactly the channel your model gives the S&P 500 inputs. It also supports your timing design: US news released after Dublin closes reaches the ISEQ the next day (your trading-hours figure).
Where it goes	2.3.1 and 2.4 (US channel); Chapter 3 (why S&P 500 inputs, alignment as-of day t); H2a discussion.
Check before citing	No abstract in the metadata. Read it to confirm the direction and size of the effect before citing. Don Bredin is at UCD (Smurfit) — relevant to your UCD options.

A6 Sharkasi, Ruskin & Crane (2006) — Long-term memory in the Irish market (ISEQ): evidence from wavelet analysis

Venue not in metadata · Basis [A] · Priority ★★ · Strand 1 (Semantic Scholar)

What it does	Applies R/S, modified R/S, GPH and discrete wavelet tests to daily returns of five ISEQ indices (Overall, Financial, General, Small Cap, ITEQ), and to absolute and squared returns.
What it finds	No long memory in returns, but strong long memory in squared and absolute returns — i.e. persistent volatility in the ISEQ.
Why it matters to you	ISEQ-specific evidence for the stylised fact behind HAR (built to approximate long memory with daily/weekly/monthly components) and behind giving the LSTM a 22-day look-back. Matches your own volatility-clustering figure (returns barely autocorrelated; squared returns strongly autocorrelated).
Where it goes	2.5 (stylised facts); Chapter 3 (HAR components; look-back choice).
Check before citing	Find the venue (likely a conference or working paper) and cite accordingly; the data end date is not in the abstract.

A7 Mitra & Duggar (2007) / Duggar & Mitra (2009) — External linkages and contagion risk in Irish banks

IMF Working Paper 07/44 (doi 10.5089/9781451866087.001) → IMF Staff Papers 2009 (doi 10.1057/imfsp.2008.38) · Basis [A] (working paper) · Priority ★★

What it does	Uses distance-to-default series built from Irish bank share prices (daily, Jan 1994 – Nov 2005): rolling correlations with banks in other European countries and the US, then a multinomial logit of “co-exceedances” (large shocks on the same day), controlling for Irish and global factors.
What it finds	Cross-border interdependence of Irish banks rose over time, differed pre-euro, post-euro and post-9/11, and there was significant contagion risk from the UK, the US and the Netherlands .
Why it matters to you	Pre-crisis warning that Irish banks — a large part of the ISEQ before 2008 — were exposed to US and UK shocks. It supports your US channel (H2a) and your FTSE robustness run. The co-exceedance idea (joint extreme days) is a useful contrast to your approach: they study tails; you study how a forecaster uses each channel.
Where it goes	2.3.2 (GFC and Irish banks); 2.4; H2a discussion.
Check before citing	Two versions of one paper: cite the 2009 IMF Staff Papers version; the abstract read here is the 2007 working paper.

A8 Bentes, Ferreira & Menezes (2013) — A FIGARCH approach to stock market volatility: Portugal, Ireland, Italy, Greece and Spain

doi 10.7813/2075-4124.2013/5-6/a.14 · Basis [T] · Priority ★ · Strand 1 (Semantic Scholar)

What it does	From the title: fractionally integrated GARCH (long-memory volatility) fitted to the five peripheral euro-area markets, Ireland included.
Why it matters to you	A second long-memory source that includes the ISEQ in a crisis-era sample; supports the HAR/long-memory argument (with A6).
Where it goes	2.5 (long memory); Chapter 6 (FIGARCH as an alternative benchmark you did not fit).
Check before citing	Low-visibility journal; no abstract. Use only as supporting evidence after reading.

A9 Corbet & Twomey (2014 SSRN; published 2015) — European equity market contagion: an empirical application to Ireland's sovereign debt crisis

SSRN doi 10.2139/ssrn.2455326 → European Financial and Accounting Journal 2015(3) (per your Handbook) · Basis [A][H] · Priority ★★★

What it does	Dynamic conditional correlation (DCC) multivariate GARCH on daily European equity returns during the Irish crisis: did the collapse of Irish equities and the troika intervention spill over to other European markets?
What it finds	Strong contagion between Irish and the other European markets during 2007–2010, which eased sharply after the troika intervention ; the authors read this as support for bail-outs as a way to absorb contagion from a single state's crisis.
Why it matters to you	Your Irish-crisis window (23 Apr 2010 – 26 Jul 2012) is mostly after the November 2010 troika programme. If contagion eased after the bail-out, that is one candidate explanation for your H2b result: the euro share rose only modestly (0.211 → 0.236) and the domestic share did not rise (0.512). Direction matters: this paper looks at Ireland → Europe; your model measures what foreign inputs contribute to forecasts of Irish volatility.
Where it goes	2.3.3 (Irish sovereign crisis); 2.4; H2b discussion; PhD approach to Twomey and Corbet.
Against your results	Partly consistent (contagion concentrated early; muted later).
Check before citing	Cite the 2015 journal version; check its exact window dates against yours.

A10 Corbet & Twomey (2014) — How have contracts for difference affected Irish equity market volatility?

SSRN doi 10.2139/ssrn.2395337 · Basis [A] · Priority ★ · Strand 1 (Semantic Scholar)

What it does	EGARCH models of Irish stock volatility before and after contracts for difference (CFDs) were introduced.
What it finds	CFDs appear to have lowered stock-specific volatility on the Irish Stock Exchange, contrary to the usual worry about leveraged products; the proposed mechanism is bid/ask “overhangs” created by CFD brokers' hedging.
Why it matters to you	Background on Irish market microstructure and a reminder that ISEQ volatility also has domestic, institutional drivers — relevant to why the domestic channel stays the largest in every window (≈ 0.48 – 0.54).
Where it goes	2.3.1 (context, one sentence at most).
Check before citing	Check whether it was later published in a journal.

A11 Kelly & Ó Gráda (2000) — Market contagion: evidence from the panics of 1854 and 1857

American Economic Review · doi 10.1257/aer.90.5.1110 · Basis [A] · Priority ★★ · Strand 1 (Crossref) · ABDC A* · 210 citations

What it does	Tests a model of contagion in which bad news spreads through personal networks, using Irish immigrant depositors of a New York bank in two 1850s panics.
What it finds	Depositors' social networks — set largely by their place of origin in Ireland and where they lived in New York — were the main determinant of who ran on the bank.
Why it matters to you	A top-journal, historical study of how contagion travels through information networks — ideal for the “This Time Is Different?” framing and for your history-based personal statement. It gives Section 2.2/2.4 a mechanism (information spreading through connected participants) that your SHAP channels can be compared with, without claiming causality.
Where it goes	2.2 (why crises recur; history); 2.4 (mechanisms of contagion).
Check before citing	Morgan Kelly is also the author of “The Irish credit bubble” (2009) in your Handbook — do not confuse the two “Kelly” citations.

A12 Conlon & Cotter (2014) — Anatomy of a bail-in

Journal of Financial Stability · doi 10.1016/j.jfs.2014.04.001 · Basis [A] · Priority ★ · Strand 1 (Semantic Scholar) · 54 citations

What it does	Applies the proposed EU bank bail-in framework retrospectively to European banks that failed in the global financial crisis.
What it finds	Equity and subordinated bond holders would have borne most of the €535bn impairment losses; senior debt holders of Greek, Austrian and Irish banks would also have needed to be bailed in.
Why it matters to you	Shows how deep the Irish banking losses were relative to Europe — context for why the Irish crisis had a strong domestic component (Section 2.3.3).
Where it goes	2.3.2–2.3.3 (context).
Check before citing	Context only; not about equity volatility.

Part B — How shocks travelled in Europe (Strand 1 context)

Studies of contagion and spillovers among European markets that include Ireland. They matter most for interpreting your H2 results (which channel the model leans on in each crisis).

B1 Ahmad, Bhanumurthy & Sehgal (2014) — The Eurozone crisis and its contagion effects on the European stock markets

Studies in Economics and Finance · doi 10.1108/sef-01-2014-0001 · Basis [A] · Priority ★★ · ABDC B · 34 citations

What it does	DCC-GARCH of contagion from the GIPSI markets and the US to seven euro-area and six non-euro European markets, 2009–2012.
What it finds	Spain, Italy, Portugal and Ireland were among the most contagious GIPSI markets; France, Belgium, Austria and Germany (euro) and the UK, Sweden and Denmark (non-euro) were hit hardest.
Why it matters to you	Places Ireland as a source of contagion in the sovereign crisis. Your model looks the other way (what foreign markets add to forecasts of Irish volatility), which helps explain why H2b's "domestic share rises" need not follow even if Ireland was central to the crisis.
Where it goes	2.3.3; 2.4; H2b discussion.
Check before citing	Their period (2009–2012) overlaps but does not match your window.

B2 Dajčman (2013) — Forbes and Rigobon's contagion test with endogenously defined crisis periods (Eurozone stock markets)

Engineering Economics · doi 10.5755/j01.ee.24.4.5419 · Basis [A] · Priority ★★

What it does	Modifies the Forbes–Rigobon adjusted-correlation test with a moving window split into two halves, so crisis periods are not chosen ad hoc; applies it to France, Germany, Ireland, Italy and Spain, Dec 2003 – Jan 2012.
What it finds	Shift contagion mattered; at the start of the Greek debt crisis (May 2010) there was contagion from Ireland, Italy and Spain to France and Germany .
Why it matters to you	Methodological contrast: your crisis windows are fixed in advance (pre-registration) — a defensible choice this paper helps you discuss (ad hoc vs data-driven windows). Its sample almost matches yours at the start (Dec 2003) and gives Ireland-specific evidence for the sovereign crisis.
Where it goes	2.4 (Forbes & Rigobon in your Handbook; this is its application); Chapter 3 (window definition); H2b.
Check before citing	Read how the crisis periods were detected before comparing dates with yours.

B3 Trabelsi & Hmida (2018) — A dynamic correlation analysis of financial contagion: evidence from the Eurozone stock markets

Entrepreneurial Business and Economics Review · doi 10.15678/eber.2018.060308 · Basis [A] · Priority ★

What it does	Bivariate DCC-GARCH between Greece and Belgium, France, Portugal, Ireland, Italy and Spain in the subprime and sovereign crises.
What it finds	Contagion in all pairs during the subprime crisis, but only Greece–Portugal during the sovereign debt crisis; rating revisions had limited effect.
Why it matters to you	Evidence that sovereign-crisis contagion into Ireland was weaker than GFC contagion — consistent with your modest euro-share rise in H2b and with the US share peaking in the GFC window among the core crises.
Where it goes	2.4; H2a/H2b discussion.
Check before citing	Greece-centred pairs only; do not generalise to Ireland–Germany.

B4 Goldstein, McCarthy & Orlov (2019) — The core, periphery, and beyond: stock market comovements among EU and non-EU countries

The Financial Review · doi 10.1111/fire.12186 · Basis [A] · Priority ★★

What it does	Linear and nonlinear correlations, copulas, quantile and lower-tail dependence for the PIIGS, France, Germany and the UK, 1999–2015.
What it finds	Each peripheral market (Ireland included) co-moves more with France, Germany and the UK than with the other peripheral markets ; the structure did not change in the GFC; non-EU comovements are weaker.
Why it matters to you	Supports using the DAX (a core market) as the euro-area channel rather than a peripheral index. Also supports your FTSE robustness run (UK co-movement with Ireland) and gives a prior for Brexit (weaker UK co-movement after exit).
Where it goes	Chapter 3 (choice of DAX); 2.4; H2d.

B5 Mensi, Boubaker, Al-Yahyaee & Kang (2018) — Dynamic volatility spillovers and connectedness between global, regional and GIPSI stock markets

Finance Research Letters · doi 10.1016/j.frl.2017.10.032 · Basis [A] · Priority ★★ 158 citations

What it does	Diebold–Yilmaz spillover index (static and rolling) and network connectedness for global, regional and GIPSI markets across recent crises.
What it finds	Crisis intensify volatility spillovers; the GIPSI markets except Spain and Greece — so Ireland is a net transmitter , as are global and US markets; the rest are net receivers.
Why it matters to you	The Diebold–Yilmaz method is now your exploratory E1 (30 Sep 2026); this is a well-cited application that includes Ireland. Reminder for Chapter 5: SHAP shares show what your forecaster relies on, not who transmits to whom — a net-transmitter finding does not contradict a large domestic share.
Where it goes	2.4 (DY applications, next to Diebold & Yilmaz 2009 in your Handbook); Chapter 6 (future work).

B6 Caporale, Spagnolo & Spagnolo (2016) — Macro news and stock returns in the euro area: a VAR-GARCH-in-mean analysis

International Review of Financial Analysis · doi 10.1016/j.irfa.2016.03.016 · Basis [A] · Priority ★

What it does	VAR-GARCH-in-mean of how newspaper coverage of macro news affects stock returns in eight euro-area countries including Ireland, 1994–2013.
What it finds	News moves returns in the expected direction; news volatility lowers returns and raises volatility; reactions are larger for negative news, in the PIIGS and in the crisis.
Why it matters to you	Evidence of asymmetric, crisis-amplified reactions in peripheral markets including Ireland — context for your direction-vs-size SHAP result in the GFC.
Where it goes	2.4 (one supporting sentence).

B7 Aslam, Ferreira, Mughal & Bashir (2021) — Intraday volatility spillovers among European financial markets during COVID-19

International Journal of Financial Studies · doi 10.3390/ijfs9010005 · Basis [A] · Priority ★★ 65 citations

What it does	Diebold–Yilmaz on 5-minute data for twelve European markets, 2 Dec 2019 – 29 May 2020, split at the pandemic declaration (11 Mar 2020).
What it finds	About 78% of intraday forecast-error variance came from spillovers; Germany and the Netherlands were the largest net transmitters; Poland and Ireland transmitted least to others; spillovers were more stable after the declaration.

Why it matters to you	Direct COVID evidence covering Ireland and almost exactly your COVID window (19 Feb – 30 Jun 2020). It supports the “everything moved together” premise of H2c, which you found only narrowly (spread 0.308 vs 0.312 in calm).
Where it goes	2.3.4 (COVID); H2c discussion.
Against your results	Their German/Dutch dominance contrasts with your falling euro (DAX) share in COVID (0.180) — worth discussing (daily vs intraday data; a forecaster's attributions vs spillover shares).

B8 Carneiro, Gomes, Lopes & Pereira (2025) — Spillovers between Euronext stock indices: the COVID-19 effect

International Journal of Financial Studies · doi 10.3390/ijfs13020066 · Basis [A] · Priority ★★

What it does	ARMA-GARCH filtering then copulas for pairs of Euronext indices (the ISEQ has traded on Euronext Dublin since 2018), before vs after COVID.
What it finds	Mostly symmetric dependence; contagion only for the CAC 40 – ISEQ pair (and a dependence change for AEX/OBX); most interdependencies fell from pre- to post-pandemic.
Why it matters to you	The only paper found that names a COVID contagion link into the ISEQ , and it runs through France, not Germany. Your euro channel uses the DAX, whose share fell in COVID. That is a concrete limitation and robustness idea (CAC 40 as an alternative euro proxy).
Where it goes	2.3.4; H2c discussion; Chapter 6 (limitations / future work).
Against your results	Contrasts with your COVID result (US share rose to 0.332; euro share fell).

B9 Gołab & Zamojska (2019) — Volatility spillovers between European financial markets: evidence since the Brexit

International Journal of Business and Globalisation · doi 10.1504/ijbg.2019.102909 · Basis [A] · Priority ★

What it does	Diebold–Yilmaz (2009) spillover index, CCC-GARCH and Cholesky decomposition for European markets including Ireland, Jan 2010 – Nov 2016 (covers the sovereign crisis, Greece, China's 2015 crash and the Brexit vote).
Why it matters to you	A Brexit spillover study that includes Ireland — background for H2d (FTSE robustness run).
Where it goes	2.3/2.4 (Brexit validation event); H2d.
Check before citing	Modest journal; the abstract states no clear Ireland-specific result.

B10 Izzeldin, Muradoglu, Pappas & Sivaprasad (2026) — Brexit and its impact on EU financial markets

International Journal of Finance & Economics · doi 10.1002/ijfe.70149 · Basis [A] · Priority ★★

What it does	Builds a “Brexit intensity” score from market reactions to negotiation events and relates it to volatility spillovers across EU countries.
What it finds	Brexit events increased volatility transmission within the EU; France was the key transmitter; the UK transmitted during the early negotiation stage; among the smaller markets, Ireland , Portugal and Spain were particularly affected.
Why it matters to you	Recent, Ireland-relevant support for your H2d prediction (UK share rises around Brexit) — balances Li (2020) in your Handbook, which reported a falling UK influence.
Where it goes	2.3/2.4 (Brexit); H2d when the FTSE run finishes.

Part C — Machine learning versus GARCH and HAR (Strand 2 core)

The forecasting literature for Section 2.5 and, above all, for discussing your H1 result: the LSTM was significantly **worse** than GARCH and HAR overall (QLIKE 0.463 vs 0.380 and 0.414), best only in COVID-19 and 2022, and worst in the GFC.

C1 Audrino & Chassot (2025) — HARd to beat: the overlooked impact of rolling windows in the era of machine learning

International Journal of Forecasting · doi 10.1016/j.ijforecast.2025.06.003 (arXiv 2024) · Basis [A] · Priority ★★★

What it does	Compares HAR with tuned machine-learning models on realised volatility for 1,455 stocks, focusing on the fitting scheme: training-window length and how often models are re-estimated. Evaluation with QLIKE, MSE and realised utility.
What it finds	With a well-chosen fitting scheme, HAR is not beaten by ML , even after extensive tuning, when both use only realised volatility and VIX; HAR is also interpretable and far cheaper to run.
Why it matters to you	The strongest published support for your H1 outcome. It lets Chapter 5 say your result sits within a live debate rather than being an anomaly. It also speaks to your design: your models use an expanding window with annual refits . Audrino & Chassot show that this choice alone can change the ranking — a limitation and a robustness idea (rolling windows).
Where it goes	2.5; Chapter 3 (walk-forward design); Chapter 5 (H1); Chapter 6.
Against your results	Consistent with your result.
Check before citing	Check whether they use daily or 5-day horizons and whether their QLIKE is on variance or log variance.

C2 Kılıç (2025) — Linear and nonlinear econometric models against machine learning models: realized volatility prediction

Federal Reserve Board, Finance and Economics Discussion Series 2025-061 · doi 10.17016/feds.2025.061 · Basis [A] · Priority ★★★

What it does	S&P 500 realised volatility, rolling forecasts from 2006: ARFIMA, HAR, regime-switching HAR (threshold, smooth-transition, Markov) versus XGBoost, feed-forward nets and recurrent nets (LSTM, LSTM with attention, GRU, BRNN).
What it finds	Regime-switching HAR models beat both ML and linear models , especially when predictors are few; ML captures some nonlinearity but gives no consistent advantage over simpler, interpretable models.
Why it matters to you	Your LSTM has only six inputs — the “few predictors” case where Kılıç finds ML weakest. Regime switching is the econometric version of your crisis story: your LSTM failed in the GFC (QLIKE 1.021 vs GARCH 0.401). A regime-switching HAR is a natural future-work benchmark.
Where it goes	2.5; Chapter 5 (H1, H3); Chapter 6.
Against your results	Consistent with your result.
Check before citing	A discussion paper, not peer-reviewed — cite as such.

C3 Rahimikia & Poon — Machine learning for realised volatility forecasting

SSRN 2020 (doi 10.2139/ssrn.3707796) and Journal of Empirical Finance (doi 10.1016/j.jempfin.2026.101739) · Basis [A] (both versions) · Priority ★★★

What it does	Compares ML with the HAR family for NASDAQ stocks (2007–2016) using limit-order-book (LOBSTER) data and Dow Jones news; the journal version uses 146 variables and SHAP .
What it finds	ML beats HAR on normal days (about 90% of the test period) but there is a persistent trade-off: models good on normal days do badly on jump days and vice versa. With SHAP, simple limit-order-book prices drive the forecasts; jumps need larger models and idiosyncratic variables.

Why it matters to you	Mirrors your pattern closely: the LSTM is near the benchmarks in calm periods (0.367 vs 0.351/0.360) but fails when volatility jumps (GFC; the frozen GFC model at $4.26 \times \text{HAR}$). A published precedent for “fine on normal days, fails on jumps”. One of the few RV papers that uses SHAP — useful in 2.6 for how others use explanations in this exact setting.
Where it goes	2.5; 2.6; Chapter 5 (H1 by window, H3).
Against your results	Consistent (normal vs jump days).
Check before citing	Cite the journal version; its DOI suggests 2026 publication — confirm the year and volume.

C4 Jia & Yang (2021) — Forecasting volatility of stock index: deep learning model with likelihood-based loss function

Complexity · doi 10.1155/2021/5511802 · Basis [A] · Priority ★★★

What it does	Trains DNN and LSTM volatility models with a likelihood-based loss instead of a distance loss (such as MSE) and evaluates all models by the test-sample likelihood.
What it finds	Distance losses introduce error (the target is itself an estimate) and make comparisons with econometric models unfair; with the likelihood loss, the deep models beat the econometric model and the LSTM does best.
Why it matters to you	Your LSTM is trained with MSE on log realised variance but judged on QLIKE , whereas GARCH is fitted by likelihood. This paper gives a published reason why that mismatch could handicap the LSTM — a candidate explanation for H1 to discuss (not a proven cause). Clear future-work step: train the LSTM on a QLIKE-type loss.
Where it goes	Chapter 3 (loss choice); Chapter 5 (H1); Chapter 6.
Check before citing	Complexity is a lower-ranked outlet; pair it with Patton (2011) from your Handbook when discussing loss functions.

C5 Bucci (2020) — Realized volatility forecasting with neural networks

Journal of Financial Econometrics 18(3) · doi 10.1093/jfinec/nbaa008 · Basis [A][H] · Priority ★★(already carded)

What it finds	Recurrent networks (LSTM, NARX) outperform the traditional econometric models, and capturing long-range dependence helps even in a highly volatile period.
Why it matters to you	The main “for” citation behind H1 in your hypothesis table. Your Handbook notes it uses monthly S&P 500 data (1950–2017) — one clear difference from your daily, single-market, shorter sample when you explain why your result differs.
Where it goes	2.5; 2.7 (H1 derivation); Chapter 5.
Against your results	Against your result.

C6 Christensen, Siggaard & Veliyev (2023) — A machine learning approach to volatility forecasting

Journal of Financial Econometrics 21(5) · doi 10.1093/jfinec/nbac020 · Basis [A][H] · Priority ★★(already carded)

What it finds	With little tuning, ML (regularisation, trees, neural nets) is competitive with and beats the HAR family on Dow Jones constituents, even with only HAR's lags as inputs; gains grow with the horizon; ML is better at using extra predictors. Their accumulated-local-effects importance shows agreement on the most important predictors but disagreement on their ranking .
Why it matters to you	The other main “for” citation for H1 — note the differences: many stocks, intraday realised variance, more predictors. The ranking-disagreement point is a direct parallel to your H4 result (channel rankings unstable across seeds in the GFC and Irish crisis: Spearman 0.7).
Where it goes	2.5; 2.6; Chapter 5 (H1, H4).

Against your results	Against on H1; consistent on H4.
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C7 Zhang, Zhang, Cucuringu & Qian (2023) — Volatility forecasting with machine learning and intraday commonality

Journal of Financial Econometrics · doi 10.1093/jffinec/nbad005 (SSRN 2022 version also found) · Basis [A] · Priority ★★

What it does	Forecasts intraday realised volatility with ML by pooling many stocks and adding a market-volatility proxy.
What it finds	Neural networks beat linear and tree models by capturing latent interactions; results hold for stocks not in the training set (a “universal” volatility mechanism).
Why it matters to you	Shows where neural networks win: large pooled panels and intraday data. Your LSTM learns from one market with about 1,000 days in the first training window — a data-size argument for Chapter 5 and a design idea (pooling several European indices) for Chapter 6/PhD.
Where it goes	2.5; Chapter 5 (H1); Chapter 6.
Against your results	Against, but under very different data conditions.

C8 Kim & Won (2018) — Forecasting the volatility of stock price index: a hybrid LSTM with multiple GARCH-type models

Expert Systems with Applications · doi 10.1016/j.eswa.2018.03.002 · Basis [A][H] · Priority ★★· 783 citations

What it does	KOSPI 200: hybrids that feed one to three GARCH-type forecasts into an LSTM, compared with GARCH, EGARCH, EWMA, a deep feed-forward net, a plain LSTM and DFN hybrids.
What it finds	The LSTM with three GARCH-type inputs (GEW-LSTM) had the lowest errors on MAE, MSE and heteroscedasticity-adjusted measures, e.g. MAE 37% below the best existing hybrid.
Why it matters to you	The most-cited LSTM–GARCH paper. It suggests that a hybrid (GARCH forecasts as LSTM inputs) may beat both — a strong Chapter 6 idea, given GARCH was your best single model.
Where it goes	2.5; Chapter 6.
Against your results	Against (for a hybrid, not a plain LSTM).
Check before citing	Evaluated on MAE/MSE, not QLIKE; a single market.

C9 Liu (2019) — Novel volatility forecasting using deep learning LSTM recurrent neural networks

Expert Systems with Applications · doi 10.1016/j.eswa.2019.04.038 · Basis [A] · Priority ★· 214 citations

What it finds	For the S&P 500 and Apple, LSTM performed as well as support-vector regression for longer-interval volatility and both beat GARCH.
Why it matters to you	A widely cited “LSTM beats GARCH” result for your literature table — useful to show the optimism you tested.
Where it goes	2.5.
Against your results	Against your result.

C10 Taneva-Angelova & Granchev (2025) — Deep learning and Transformer architectures for volatility forecasting: evidence from US equity indices

Journal of Risk and Financial Management 18(12) · doi 10.3390/jrfm18120685 · Basis [A] · Priority ★★

What it does	HAR-RV, ARIMA and GARCH versus LSTM, CNN-LSTM, PatchTST-lite and a Transformer for S&P 500, NASDAQ 100 and Dow Jones realised variance, 2000–2025, horizons of 1, 5 and 22 days , evaluated with QLIKE , RMSE, MAE and Diebold–Mariano tests.
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What it finds	Transformer-based models had the lowest errors, especially at short horizons and in volatile periods.
Why it matters to you	The closest match to your evaluation toolkit (QLIKE + DM at a 5-day horizon) — a direct comparator for Chapters 3 and 5. Their “better in volatile periods” finding fits your COVID and 2022 windows (LSTM best) but not the GFC.
Where it goes	2.5; Chapter 3; Chapter 5.
Against your results	Mixed: against overall, consistent for COVID/2022.

C11 Chung Alva, Espinoza Poves & Quispe Llanos (2025) — Forecasting financial volatility under structural breaks: GARCH models and deep learning

Journal of Risk and Financial Management 18(9) · doi 10.3390/jrfm18090494 · Basis [A] · Priority ★★

What it does	Four Latin American indices, 2000–2024: GARCH (with breaks found by a modified ICSS algorithm) versus LSTM and CNN.
What it finds	Ignoring breaks overstates volatility persistence and hurts GARCH; allowing for breaks helps GARCH only sometimes; deep learning beats GARCH at medium and long horizons.
Why it matters to you	Your GARCH persistence is 0.957–0.999 across refits — this paper gives the reason to comment on it (crisis breaks inflate persistence). Its crisis/structural-break framing is close to your RQ3 (Minsky) logic.
Where it goes	2.5; Chapter 5 (GARCH parameters, H3).
Against your results	Against on H1 (at longer horizons).
Check before citing	Check the author surname format (Spanish double surnames) in the published version.

C12 Souto & Moradi (2023) — Forecasting realized volatility through financial turbulence and neural networks

Economics and Business Review · doi 10.18559/ebv.2023.2.737 · Basis [A] · Priority ★★

What it does	Adds a “financial turbulence” risk metric to an LSTM and compares it with two other LSTMs, GARCH(1,1), EGARCH(1,1) and HAR.
What it finds	The turbulence-augmented LSTM was more accurate and robust for middle-to-high-volatility stocks; for low-volatility stocks only its robustness advantage was clear.
Why it matters to you	Suggests that giving the LSTM an explicit turbulence/regime signal helps in stressed markets — relevant to your GFC failure and a concrete future-work input.
Where it goes	2.5; Chapter 6.

C13 Souto & Moradi (2023) — Introducing NBEATSx to realized volatility forecasting

Expert Systems with Applications · doi 10.1016/j.eswa.2023.122802 · Basis [A] · Priority ★

What it finds	Across six indices, NBEATSx beat LSTM, TCN, HAR, GARCH and GJR-GARCH, but its accuracy advantage was not evident for developing-country indices .
Why it matters to you	A reminder that ML gains are market-dependent; useful when arguing that a small market such as the ISEQ may not behave like the large indices most studies use.
Where it goes	2.5 (one sentence).

C14 Ramos-Pérez, Alonso-González & Núñez-Velázquez (2021) — Multi-Transformer: a neural network architecture for forecasting S&P volatility

Mathematics · doi 10.3390/math9151794 · Basis [A] · Priority ★

What it finds	Hybrids built on Multi-Transformer/Transformer layers were more accurate than autoregressive models and LSTM/feed-forward hybrids for S&P volatility; motivated by the GFC and COVID losses.
Why it matters to you	Evidence that newer architectures (Transformers) may outperform LSTMs — a future-work pointer, and context for why you chose an LSTM (interpretability tooling, smaller data).
Where it goes	2.5; Chapter 6.

C15 Song, Tang, Wang & Ma (2022) — Volatility forecasting incorporating macroeconomic variables: GARCH-MIDAS and deep learning

Journal of Forecasting · doi 10.1002/for.2899 · Basis [A] · Priority ★

What it finds	Low-frequency macro variables processed through GARCH-MIDAS and fed to ML/DL models improved realised-volatility forecasts; deep learning did best.
Why it matters to you	A hybrid route that adds macro information — relevant if you discuss what your six return-based inputs leave out (e.g. Irish sovereign spreads).
Where it goes	Chapter 6 (limitations: inputs).

C16 Moreno-Pino & Zohren (2024) — DeepVol: volatility forecasting from high-frequency data with dilated causal convolutions

Quantitative Finance · doi 10.1080/14697688.2024.2387222 · Basis [A] · Priority ★

What it finds	A dilated-convolution model on NASDAQ-100 intraday data extracts information that precomputed daily realised measures lose.
Why it matters to you	Supports a limitation you should state: your realised variance is built from daily squared returns because free intraday ISEQ data are not available.
Where it goes	Chapter 3 (target construction); Chapter 6 (limitations).

C17 Luong & Dokuchaev (2018) — Forecasting of realised volatility with the random forests algorithm

Journal of Risk and Financial Management · doi 10.3390/jrfm11040061 · Basis [A] · Priority ★

What it finds	An extended HAR with purified implied volatility, estimated with random forests, improved forecasts of the direction and size of realised volatility.
Why it matters to you	An example of HAR + ML combined rather than opposed — supports treating HAR as a strong baseline to build on.
Where it goes	2.5 (one sentence).

C18 Gupta, Nel & Pierdzioch (2021) — Investor confidence and forecastability of US stock market realized volatility: machine learning

Journal of Behavioral Finance · doi 10.1080/15427560.2021.1949719 · Basis [A] · Priority ★

What it finds	Random forests on monthly US realised volatility (2001–2020): investor confidence, and especially its uncertainty, adds out-of-sample predictive value.
Why it matters to you	Sentiment/confidence as an omitted input — a limitation/future-work point, not central.
Where it goes	Chapter 6.

C19 Chun, Cho & Ryu (2024) — Volatility forecasting and volatility-timing strategies: a machine learning approach

Research in International Business and Finance · doi 10.1016/j.ribaf.2024.102723 · Basis [A] · Priority ★

What it finds	ML volatility forecasts beat standard volatility models and improved volatility-timing portfolios, especially in volatile markets.
Why it matters to you	You evaluate statistical accuracy only (QLIKE, MSE). This is the reference for adding economic value (portfolio or risk-management tests) as future work.
Where it goes	Chapter 6.

C20 Niu, Wang & Zhang (2023) — Forecasting stock market volatility with various geopolitical risk categories: machine learning evidence

International Review of Financial Analysis · doi 10.1016/j.irfa.2023.102738 · Basis [T] · Priority ★ 62 citations

What it does	From the title: tests whether categories of geopolitical risk improve ML forecasts of stock-market volatility.
Why it matters to you	The only 2022-relevant (war/geopolitics) forecasting paper the searches found. Useful for your 2022 validation window, where H2e failed (euro share fell to 0.143; US rose to 0.317).
Where it goes	2.3 (2022 shock, next to Boungou & Yatié in your Handbook); H2e discussion.
Check before citing	No abstract — read before citing.

C21 Lin et al. (2022) — Forecasting the realized volatility of stock price index: a hybrid model integrating CEEMDAN and LSTM

Expert Systems with Applications · doi 10.1016/j.eswa.2022.117736 · Basis [T] · Priority ★ 129 citations

Why it matters to you	A well-cited decomposition + LSTM hybrid for index realised volatility — representative of the “hybrid LSTM” branch in 2.5.
Where it goes	2.5 (literature table).
Check before citing	No abstract — read before citing results.

C22 Zhu, Bai, He & Liu (2023) — Forecasting realized volatility with machine learning: panel data perspective

Journal of Empirical Finance · doi 10.1016/j.jempfin.2023.07.003 · Basis [T] · Priority ★ 51 citations

Why it matters to you	With Zhang et al. (C7), evidence that pooling many series is where ML gains come from — supports the data-size argument in Chapter 5.
Where it goes	2.5; Chapter 5.
Check before citing	No abstract — read before citing results.

C23 Gu, Kelly & Xiu (2020) — Empirical asset pricing via machine learning

Review of Financial Studies · doi 10.1093/rfs/hhaa009 · Basis [A] · Priority ★ 2,520 citations

What it finds	Trees and neural networks give the largest gains in predicting returns by capturing nonlinear interactions; all methods agree on the main signals, which include momentum, liquidity and volatility.
Why it matters to you	The standard citation for “machine learning in finance works because of nonlinear interactions” — one sentence of background in 2.5 before narrowing to volatility.

Where it goes	2.5 (opening).
Check before citing	About returns, not volatility — do not cite it as volatility evidence.

Part D — Review articles (to structure Section 2.5 and support the gap)

D1 Gunnarsson, Isern, Kaloudis, Risstad, et al. (2024) — Prediction of realized volatility and implied volatility indices using AI and machine learning: a review

International Review of Financial Analysis · doi 10.1016/j.irfa.2024.103221 · Basis [A] · Priority ★★★ · 70 citations

What it does	Systematic review of AI/ML studies forecasting realised volatility and implied-volatility indices.
What it finds	AI/ML is promising and often matches or beats econometric models; memory networks (LSTM, GRU) are consistently among the best; econometric models remain highly relevant with similar results; hybrids are promising; very few papers use explainable AI , and the authors call for more XAI.
Why it matters to you	The single best citation for your research gap : a 2024 systematic review stating that XAI is rarely applied in volatility forecasting. Its balanced verdict (“econometric models commonly yield similar results”) also frames your H1 result.
Where it goes	2.5 (overview); 2.6; 2.7 (gap) ; Chapter 5.

D2 Ge, Lalbakhsh, Isai, Lenskiy & Suominen (2022) — Neural network-based financial volatility forecasting: a systematic review

ACM Computing Surveys · doi 10.1145/3483596 · Basis [A] · Priority ★★ · 60 citations

What it finds	Reviewing 35 studies published after 2015, it finds results are hard to compare across studies and there is a large gap between modern ML and the models applied to volatility; it proposes a shared evaluation task.
Why it matters to you	Supports your emphasis on a pre-registered, like-for-like evaluation (same windows, same loss, DM tests) — a methodological strength to name in Chapter 3.
Where it goes	2.5; Chapter 3.

D3 Qiu, Kownatzki, Scalzo & Cha (2025) — Historical perspectives in volatility forecasting methods with machine learning

Risks 13(5) · doi 10.3390/risks13050098 · Basis [A] · Priority ★★

What it finds	A survey of how volatility forecasting evolved, with a benchmark of landmark models (implied volatility, GARCH, LSTM, Transformer) and open-source benchmark code.
Why it matters to you	A history-of-methods narrative that fits your “This Time Is Different?” framing and gives Section 2.5 a clean chronological spine (ARCH → GARCH → HAR → LSTM → Transformers).
Where it goes	2.5 (structure).

D4 Leushuis & Petkov (2026) — Advances in forecasting realized volatility: a review of methodologies

Financial Innovation · doi 10.1186/s40854-025-00809-5 · Basis [A] · Priority ★

What it finds	Reviews realised-volatility models in the academic literature from 2000 to mid-2024; a hybrid CNN-LSTM is the top performer in their comparison.
Why it matters to you	The most recent survey — cite it to show your review is current; its hybrid finding supports the Chapter 6 hybrid idea.
Where it goes	2.5; Chapter 6.

Part E — Explainability (SHAP) in volatility forecasting

The literature for Section 2.6 and for defending how you grouped SHAP attributions. **Read E1 and E2 before you claim novelty.**

E1 Choi, Shin & Shin (2024) — Vector SHAP values for machine learning time series forecasting

Journal of Forecasting · doi 10.1002/for.3220 · Basis [A] · Priority ★★★ · 30 citations

What it does	Proposes vector SHAP : instead of one Shapley value per lag of each variable, it gives one value per variable's whole vector of lags. Proves properties (vector local accuracy, missingness, consistency) and runs a Monte Carlo comparison with standard and sampling SHAP.
What it finds	Vector SHAP is much faster, close to summed standard SHAP, and less sensitive to the sampling proportion that makes sampling SHAP unstable. Applied to forecasting Korea's KOSPI realised volatility from the realised volatility of 16 national indices , and grouping them by region (Europe, North America, Asia) gives regional values close to the sum of the country values.
Why it matters to you	This is the closest methodological precedent to your RQ2. You group GradientShap attributions over lags and inputs into channels (domestic, US, euro) and compare channel shares; they group lags into variables and variables into regions to explain a domestic RV forecast. It helps you in three ways: (i) it justifies grouping as a principled step; (ii) its finding that grouped values $\approx$ sums of individual values supports your summing approach; (iii) its point about sampling sensitivity bears on your H4 instability (0.7 in the GFC/Irish windows). It also narrows your novelty claim: grouping foreign-market SHAP into regions for a domestic RV forecast is not new . What remains distinctive in your study: Ireland, crisis-by-crisis fingerprints with pre-registered hypotheses by crisis origin, seed stability, and the frozen-model (Minsky) test.
Where it goes	2.6; 2.7 (position your contribution against it); Chapter 3 (grouping method); Chapter 5 (H4).
Check before citing	Check whether they use absolute or signed values when grouping — you use shares of absolute attribution.

E2 Shin, Choi & Shin (2025) — A comparison of vector SHAP explanations of the S&P 500 realized volatility forecast by LSTM and ADL models

Korean Journal of Applied Statistics 38(3) · doi 10.5351/kjas.2025.38.3.485 · Basis [A] · Priority ★★★

What it does	Applies vector SHAP to S&P 500 realised-volatility forecasts from an LSTM and an autoregressive distributed-lag (ADL) model, with realised volatility, implied volatility and search volume (and their lags) as predictors.
What it finds	Past realised volatility contributes most; search volume adds little; in the ADL model multicollinearity makes contributions hard to read.
Why it matters to you	The nearest paper to your design: LSTM + SHAP on realised volatility . Its finding that the market's own past volatility dominates parallels your large domestic share (≈ 0.48 – 0.54 in every window). What it does not do (from the abstract): crises, cross-market channels, Ireland — the gap your study fills.
Where it goes	2.6; 2.7; Chapter 5 (domestic dominance).
Check before citing	Published in Korean with an English abstract — read what you can and cite with care.

E3 Kang, Ryu & Webb (2026) — Uncertainty indicators as key predictors of oil volatility: an interpretable machine learning approach

Computational Economics · doi 10.1007/s10614-025-11299-z · Basis [A] · Priority ★★

What it does	Monthly crude-oil volatility 1986–2023 with 140 macro and uncertainty variables; tree models with time-varying interpretability across recessions.
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What it finds	Uncertainty indicators, not oil fundamentals, are the key predictors; the tree models' advantage varies by regime, and time-varying interpretability reveals what static importance scores miss.
Why it matters to you	Direct support for your core idea that explanations should be read by regime (crisis window) , not averaged over the whole sample.
Where it goes	2.6; Chapter 3 (why SHAP per window).

E4 Hashami & Maldonado (2025) — Can news predict the direction of oil price volatility? A language-model approach with SHAP explanations

arXiv 2508.20707 · Basis [A] · Priority ★

What it finds	Most sentiment indicators do not consistently beat HAR; the raw news count is a robust predictor; word-level SHAP drivers shift across regimes (pre-pandemic, early pandemic, post-shock).
Why it matters to you	Another example of SHAP drivers changing across regimes, including COVID, and of HAR being hard to beat.
Where it goes	2.6 (one sentence).
Check before citing	Preprint, not peer-reviewed.

E5 Zhu, Zheng, Wang, Huang & Feng (2025) — Forecasting China's precious metal futures volatility: GBRT models and a time-model combination of Tree SHAP

International Review of Financial Analysis · doi 10.1016/j.irfa.2025.104249 · Basis [T] · Priority ★

Why it matters to you	From the title: combines SHAP explanations across time and across models — relevant to how you average attributions across seeds and days.
Where it goes	2.6 (optional).
Check before citing	No abstract — read before citing.

7. Reading plan

The ten ★★★papers first — they change what you write. Rough times assume you read the introduction, method summary and results tables carefully.

#	Paper	Why now	Time
1	Choi, Shin & Shin (2024) — vector SHAP (E1)	Closest precedent for your grouped SHAP; sets the novelty boundary	2 h
2	Shin, Choi & Shin (2025) — LSTM + vector SHAP on S&P 500 RV (E2)	Nearest design to yours	1 h
3	Gunnarsson et al. (2024) — review (D1)	Evidence for the gap; balanced H1 framing	2 h
4	Audrino & Chassot (2025) — HARd to beat (C1)	Main support for your H1 result; fitting-scheme argument	2 h
5	Kılıç (2025) — econometricvs ML (C2)	Few predictors + regimes: your situation	1.5 h
6	Rahimikia & Poon — ML for RV (C3)	Normal vs jump days = your calm vs GFC pattern; uses SHAP	1.5 h
7	Jia & Yang (2021) — likelihood loss (C4)	Training-loss explanation to examine for H1	1 h
8	Gallagher & Twomey (1998) (A2)	Irish spillover origin question; fix the citation	1.5 h
9	Corbet & Twomey (2015) (A9)	Troika timing vs your H2b window	1 h
10	Bredin, Gavin & O'Reilly (2005) (A5)	The US channel into Irish volatility	1 h

Then the ★★cards in the order of your chapters (2.3 → 2.4 → 2.5 → 2.6). Add each paper to the synthesis matrix in your Blueprint as you read.

8. What the search did not cover — next steps

- **Re-run Strand 1** now that OpenAlex search works again; it ran without OpenAlex.
- **Rule these out before claiming the gap** (both matched “ISEQ LSTM” in a targeted OpenAlex check on 28 Sep 2026; neither search strand was designed to find Ireland + deep learning together): “Worldwide financial markets reactions to the Covid-19 epidemic situation: a hybrid FIEGARCH–deep learning approach” (2023, HAL working paper) and “Enhancing CNN-LSTM models for financial time-series forecasting through hyperparameter optimization and regularization” (2025, NORMA). Check whether they model the ISEQ, and what they find. Also run “ISEQ” + “machine learning” in Google Scholar once.
- **Screen the 48 + 182 records** in the app (Screening page → Include/Exclude), then export the workbook — your decisions are the real screening step.
- **Citation chasing (snowballing)**: start from Gallagher & Twomey (1998), Corbet & Twomey (2015), Bucci (2020), Christensen et al. (2023) and Choi et al. (2024) in Inciteful or Connected Papers — keyword searches miss papers that cite them.
- **Add by hand**: Vortelinos (2017), “Forecasting realized volatility: HAR against principal components combining, neural networks and GARCH” (Research in International Business and Finance) — found in the first Strand 2 run, missed in the second.
- **Ireland in 2022**: no Irish-specific study of the war/energy shock was found; check Central Bank of Ireland publications (Financial Stability Reviews).
- **DBS library databases**: if the library gives you Scopus or Web of Science web access, run the same queries there, export CSV and import it into the app (it maps their column names); that widens coverage beyond open databases.
- **Journal quality**: the app's ABDC lookup is best-effort. Check ABDC/AJG ranks yourself for anything you rely on heavily.
- **Found outside the searches (30 Sep 2026) — must cite**: Feng, Shi & Kutan (2026, Journal of International Money and Finance 160) on interpretable ML for cross-market volatility forecasting, and Brini (2026, arXiv preprint) on foundation models vs HAR. Cards: Literature Handbook v2, D5 and D4.
- **Strand 3 (exploratory extensions) not yet searched**: spillover/connectedness + SHAP or interpretable ML; low volatility / volatility paradox + banking crisis. Query strings: Chapter 2 Blueprint, Section 3.
- **Time-varying connectedness**: Irish finance groups now often use TVP-VAR connectedness (for example a 2026 SSRN paper by Conlon, Corbet & Muñiz on weekday patterns in volatility spillovers — listing seen, abstract not read). Look for Ireland-specific applications when you plan the PhD proposal.

9. Facts for your search-strategy paragraph and AI-use log

Facts only — write the paragraph yourself.

- Date: 28 September 2026. Tool: LitReviewKit (open source), run locally with the changes listed in Section 2.
- Databases: Crossref, OpenAlex, Semantic Scholar (public APIs). Not searched: Scopus, Web of Science, SSRN/RePEc directly.
- Strand 1: 9 queries, 1995–2026; Strand 2: 9 queries, 2010–2026 (terms in Section 2).
- Automatic steps: DOI/title-year de-duplication; keyword screening on title, abstract and keywords; relevance threshold 30.
- Counts: Strand 1 219 → 146 → 48; Strand 2 1,440 → 1,076 → 182 (Figure 1).
- Limitations to state: open databases only; keyword screening misses papers without abstracts; OpenAlex unavailable for Strand 1; Strand 2 rules revised after a first run; screening decisions still to be made by you.

Suggested AI-use log entry (added to notes/ai_use.md — edit it if you disagree):

Date	Tool	What it helped with	What I did myself
28 Sep 2026	Claude (Claude Code)	Set up LitReviewKit locally, wrote the two search presets, fixed its Semantic Scholar limit, ran the searches, and wrote this reading guide from abstracts and metadata	To screen the records, read the papers and write all dissertation text myself

10. Corrections for your existing documents

- **Gallagher, L. A. & Twomey, C. (1998)** — your Handbook v2 and Blueprint list it as “Twomey, C. (1998)”. Verify and correct.
- **Kearney (1998)** — search metadata says 1997; the journal volume is 1998 (Journal of Financial Research 21(1)). Keep 1998.
- **Two “Kelly” papers:** Kelly (2009) “The Irish credit bubble” (in your Handbook) and Kelly & Ó Gráda (2000) AER (new). Keep them distinct.
- **Cite journal versions, not working papers:** Duggar & Mitra (2009) not Mitra & Duggar (2007); Rahimikia & Poon (Journal of Empirical Finance) not SSRN; Zhang et al. (2023, JFEC) not SSRN (2022); Corbet & Twomey (2015) not SSRN (2014); Audrino & Chassot (2025, IJF) not arXiv (2024).
- **Status 30 Sep 2026 — applied:** Gallagher & Twomey (1998) corrected in the Literature Handbook v2 (card and sources), the Chapter 2 Blueprint (2.3 table, hypothesis table, references) and the Research Plan (claim 27, reading list 8.6). Kearney (1998) was already correct. None of the working-paper versions listed above appear in the existing documents; use the journal versions when you add these papers.